IMMERSIONS OF GROUP SCHEMES OVER A FIELD ARE CLOSED IMMERSIONS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lemma morphisms-lemma-fpqc-quotient-topology (canonical tag 02JY). cited at proof lines 1106-1110. Closedness descends along the flat qcqs surjective base change.
- Lemma lemma-reduced-subgroup-scheme-perfect (canonical tag 047R). cited at proof lines 1113-1114. Reductions remain subgroup schemes over a perfect field.
- Lemma varieties-lemma-closure-of-product (canonical tag 047B). cited at proof lines 1116-1119. The closure of the product image.
- Lemma lemma-open-subgroup-closed-over-field (canonical tag 047S). cited at proof lines 1126-1128. Open subgroup schemes over a field are closed.

Source reference. Original lemma 02JY: sources/stacks-chow/tags.txt, source lines 3353–3353. Cited at proof lines 1106-1110 to descend closedness from the perfect extension; load-bearing.

Source reference. Original lemma 047R: sources/stacks-chow/tags.txt, source lines 5518–5518. Cited at proof lines 1113-1114 to justify passing to the reductions; load-bearing.

Source reference. Original lemma 047B: sources/stacks-chow/tags.txt, source lines 5502–5502. Cited at proof lines 1116-1119 to identify the closure of the product image; load-bearing.

Source reference. Original lemma 047S: sources/stacks-chow/tags.txt, source lines 5519–5519. Cited at proof lines 1126-1128 to see i(G') is closed inside the closure, hence equal to it; load-bearing.

Lemma 0.2. *Let $i : G' \to G$ be an immersion of group schemes over a field k. Then i is a closed immersion, i.e., $i(G')$ is a closed subgroup scheme of G.*

Proof. To show that i is a closed immersion it suffices to show that $i(G')$ is a closed subset of G. Let $k \subset k'$ be a perfect extension of k. If $i(G'_{k'}) \subset G_{k'}$ is closed,

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then $i(G') \subset G$ is closed by Morphisms, Lemma 02JY (as $G_{k'} \to G$ is flat, quasi-compact and surjective). Hence we may and do assume k is perfect. We will use without further mention that products of reduced schemes over k are reduced. We may replace G' and G by their reductions, see Lemma 047R. Let $\overline{G'} \subset G$ be the closure of $i(G')$ viewed as a reduced closed subscheme. By Varieties, Lemma 047B we conclude that $\overline{G'} \times_k \overline{G'}$ is the closure of the image of $G' \times_k G' \to G \times_k G$. Hence

$$m\Big(\overline{G'} \times_k \overline{G'}\Big) \subset \overline{G'}$$

as m is continuous. It follows that $\overline{G'} \subset G$ is a (reduced) closed subgroup scheme. By Lemma 047S we see that $i(G') \subset \overline{G'}$ is also closed which implies that $i(G') = \overline{G'}$ as desired. $\square$